Question	Answer	Marks	AO	Guidance
4	**DR** e.g. $\frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)}$	**M1**	**3.1a**	Rationalising denominators At least one example of multiplying by e.g. $\frac{\sqrt{2}-1}{\sqrt{2}-1}$
	$\frac{\sqrt{2}-1}{2-1}+\frac{\sqrt{3}-\sqrt{2}}{3-2}+\frac{2-\sqrt{3}}{4-3}$ $2-1=1$	**A1**	**1.1**	Convincing completion including all steps including the denominators of 1 or better
	Alternative method			
	$\frac{(\sqrt{3}+\sqrt{2})(2+\sqrt{3})+(\sqrt{2}+1)(2+\sqrt{3})+(\sqrt{2}+1)(\sqrt{3}+\sqrt{2})}{(\sqrt{2}+1)(\sqrt{3}+\sqrt{2})(2+\sqrt{3})}$ $\frac{\sqrt{3}+3+2\sqrt{2}+\sqrt{6}+2\sqrt{2}+\sqrt{6}+2+\sqrt{3}+\sqrt{6}+2+\sqrt{3}+\sqrt{2}}{(\sqrt{6}+2+\sqrt{3}+\sqrt{2})(2+\sqrt{3})}$ $\frac{3\sqrt{6}+4\sqrt{3}+5\sqrt{2}+7}{2\sqrt{6}+\sqrt{18}+4+2\sqrt{3}+2\sqrt{3}+3+2\sqrt{2}+\sqrt{6}}$			
	$\frac{3\sqrt{6}+4\sqrt{3}+5\sqrt{2}+7}{3\sqrt{6}+4\sqrt{3}+3\sqrt{2}+2\sqrt{2}+7}$	**M1**		Putting over a common denominator and simplifying all like terms
	$\frac{3\sqrt{6}+4\sqrt{3}+5\sqrt{2}+7}{3\sqrt{6}+4\sqrt{3}+5\sqrt{2}+7}=1$	**A1**		Convincing completion including all steps
		[2]		

Q4. This is a DR question. There are two solutions possible. The first is to rationalise the denominators. To score the M1 we want to see at least one fraction multiplied by the correct fraction $\frac{\sqrt{2}-1}{\sqrt{2}-1}$ *or* $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ *or* $\frac{2-\sqrt{3}}{2-\sqrt{3}}$.

To score the A1 it must be all correct with the denominators seen as 2-1, 3-2, 4-3 or 1,1,1. There must be something there. This leads to the final answer of 1.